

LSMPS-PINN and LSFVP-PINN: Complementary meshless operator learning strategies for solving partial differential equations

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ABSTRACT

To circumvent the computational bottlenecks and instability associated with automatic differentiation (AD) in Physics-Informed Neural Networks (PINNs), this study proposes a unified meshless framework introducing two complementary architectures: the Least-Squares Moving Particle Semi-implicit physics-informed neural networks (LSMPS-PINN) and the Least-Squares Finite-Volume Particle physics-informed neural networks (LSFVP-PINN). By shifting the derivative computation to efficient precomputed discrete operators, both methods significantly outperform conventional AD-PINNs, yet they are engineered to serve distinct computational priorities. Functioning as an efficiency-oriented solver, LSMPS-PINN leverages strong-form least-squares collocation to maximize matrix sparsity, making it the optimal choice for scenarios demanding rapid convergence and low computational cost. Conversely, operating as an accuracy-oriented solver, LSFVP-PINN adopts the integral form of the governing equations, which converts high-order differential operators into lower-order flux integrations via the divergence theorem. This formulation prioritizes physical fidelity and robustness by significantly reducing sensitivity to derivative approximation errors. Validated across linear diffusion, nonlinear Burgers, and lid-driven cavity flow benchmarks, our results demonstrate that these methods offer researchers flexible, high-performance alternatives tailored to specific requirements for computational speed or physical fidelity.

1. Introduction

Solving partial differential equations (PDEs) is fundamental in scientific computing and engineering. For decades, grid-based numerical methods such as the Finite Element Method (FEM), Finite Difference Method (FDM), and Finite Volume Method (FVM) have been the dominant tools for simulating physical phenomena[1,2]. However, these traditional methods often face significant challenges when dealing with complex geometries, high-dimensional problems, or inverse problems where data is sparse or noisy. In recent years, the rapid development of deep learning has catalyzed a paradigm shift in scientific computing. Physics-Informed Neural Networks (PINNs), originally formalized by Raissi et al [3], have emerged as a powerful mesh-free framework that embeds governing PDEs directly into the training of neural networks. By incorporating residuals of the PDE, along with boundary and initial conditions, as soft constraints in the loss function, PINNs can learn the underlying physics with or without labeled data[3].

This hybrid learning approach offers several core advantages: PINNs operate without a predefined mesh, enabling continuous field

representations[4]. They naturally fuse multi-modal information, and they scale favorably to high-dimensional problems[5,6]. These features have driven rapid global interest, with numerous reviews documenting their evolution across physics and engineering[7]. Notably, researchers have actively classified neural PDE solvers into data-driven, physics-constrained, and purely physics-driven schemes, underscoring the transition toward deep learning models that are tightly coupled with physical modeling[7].

Despite their promise, existing PINN-based solvers face inherent limitations, particularly when relying on Automatic Differentiation (AD) to compute PDE residuals (AD-PINNs). While general, AD can be computationally expensive and sensitive to the complexity of the PDE, especially for high-order derivatives[8]. Research has shown that the recursive computation of high-order spatial derivatives via AD significantly slows down training and can suffer from "gradient pathology," where unbalanced gradients between loss terms hinder convergence[8, 9]. To address this, various acceleration strategies have been proposed. For instance, discrete-time or gradient-enhanced schemes and analyses of gradient flow pathologies have been studied to mitigate training

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difficulties[9], and Radial Basis Function (RBF) based networks have been used as meshless approximations to replace costly AD derivatives in some contexts[10]. Other approaches include smoothing-kernel-based gradient operators (e.g., SK-PINN) to stabilize noisy gradients[11], and feature-enforcing schemes that inject boundary conditions as priors or hard constraints to improve convergence speed [12]. Furthermore, standard PINNs often suffer from spectral bias, a phenomenon where the network learns low-frequency components significantly faster than high-frequency details. This limitation has motivated the development of various architectural and training remedies [13].

Parallel to these developments, classical mesh-free particle methods have inspired new PINN variants. Particle-based schemes like Smoothed Particle Hydrodynamics (SPH) and the Moving Particle Semi-implicit (MPS) method are renowned for handling large deformations in a Lagrangian fashion[14–18]. The Least-Squares Moving Particle Semi-implicit (LSMPS) algorithm, in particular, utilizes a weighted least squares formulation to reconstruct spatial derivatives with arbitrary high-order accuracy and consistency[19,20]. Integrating these consistent particle operators into PINNs offers a route to robust meshless learning[20,21]. However, purely point-wise or particle-based PINNs can have difficulty enforcing global conservation laws strictly, a shortcoming that conservative/discrete-domain PINN formulations have sought to address[22,23].

Complementary to particle methods, deep finite-volume approaches demonstrate that enforcing integral balance (conservation laws) on control volumes can reduce the required derivative order and significantly improve numerical accuracy. Recent works on deep finite-volume and FV-PINN formulations showed that computing residuals via face/flux integration lowers derivative order and improves shock-capturing and conservation properties[24,25]. Su et al. specifically demonstrated faster convergence on Navier–Stokes benchmarks for FV-PINNs compared to vanilla PINNs[25]. Unified frameworks combining finite-volume schemes with deep learning have also been proposed for heterogeneous PDEs and multi-physics problems[24,26].

Motivated by these observations, this work proposes a unified research framework that introduces and compares two novel parallel strategies: the LSMPS-PINN and the LSFVP-PINN (Least Square Finite Volume Particle PINN). We posit that these two models represent a strategic trade-off between efficiency and accuracy, allowing users to select the appropriate model based on specific engineering requirements:

- (1) LSMPS-PINN (Efficiency-Oriented): This model integrates the strong-form discretization of the LSMPS method into the neural network. By replacing automatic differentiation with pre-computed particle interaction matrices, it significantly accelerates the training process while maintaining the geometric flexibility of meshless methods[20,27]. It is the optimal choice for scenarios requiring rapid simulations or handling complex topological changes where computational speed is paramount.
- (2) LSFVP-PINN (Accuracy-Oriented): This model represents a novel synthesis of LSMPS and finite-volume discretization. In this scheme, we construct small control volumes around each particle and utilize the high-precision reconstruction capability of LSMPS to compute fluxes across these virtual interfaces. The network is trained to satisfy integral balance laws (a weak/integral form) to incorporate flux interactions directly. While the flux integration process incurs higher computational cost than LSMPS-PINN, it yields significantly higher numerical precision and improved numerical stability through derivative order reduction[20,24, 25].

To validate and compare these parallel methodologies, we test AD-PINN, LSMPS-PINN, and LSFVP-PINN on three canonical PDE problems: the two-dimensional diffusion equation, the two-dimensional

viscous Burgers equation (classical analyses of Burgers-type equations appear in the literature[28,29]), and the two-dimensional incompressible lid-driven cavity flow (a standard benchmark studied by Ghia et al [30]). Our results demonstrate that while LSMPS-PINN offers superior computational efficiency, LSFVP-PINN consistently achieves the highest accuracy across all cases. In summary, this work highlights the distinct advantages of incorporating least-squares finite-volume particle integration into the PINN framework, providing a flexible roadmap for balancing accuracy and cost in next-generation neural solvers.

2. Methodology

This section delineates the mathematical foundations and implementation specifics of the three PINN frameworks investigated in this study: the conventional AD-PINN, the particle-based LSMPS-PINN, and the novel hybrid LSFVP-PINN. The core distinction among them lies in their approach to computing spatial derivatives and constructing the physics-informed loss function. Collectively, these methods represent a spectrum of solutions balancing computational efficiency with physical fidelity.

2.1. The automatic differentiation PINN (AD-PINN)

The AD-PINN represents the canonical implementation of physics-informed neural networks[9]. Its essence is to leverage the automatic differentiation (AD) capabilities of modern deep learning frameworks to compute partial derivatives of the network's output with respect to its inputs, which are then used to formulate the PDE residual.

2.1.1. Mathematical framework

Let a PDE be defined generally as

$$N[u(\mathbf{x}, t)] = 0, \mathbf{x} \in \Omega, t \in [0, T] \quad (1)$$

subject to boundary conditions (BCs) and initial conditions (ICs). A neural network $u_\theta(\mathbf{x}, t)$ parameterized by θ is trained to approximate the solution u . The physics-informed loss function is constructed as

$$L(\theta) = L_r(\theta) + L_b(\theta) + L_i(\theta) \quad (2)$$

where the residual loss L_r is computed at a set of collocation points $\{\mathbf{x}_i, t_i\}_{i=1}^{N_r}$ within the domain

$$L_r(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \|N[u_\theta(\mathbf{x}_i, t_i)]\|^2 \quad (3)$$

The $BC(L_b)$ and $IC(L_i)$ losses are computed similarly at boundary and initial points. The critical feature is that the operator $N[\cdot]$, which contains all spatial and temporal derivatives, is evaluated using AD. For example, to compute the Laplacian Δu , the network is differentiated twice

$$\Delta u_\theta \approx \frac{\partial^2 u_\theta}{\partial x^2} + \frac{\partial^2 u_\theta}{\partial y^2} \quad (4)$$

where derivatives are obtained via the chain rule through the network's computational graph.

2.1.2. Implementation and characteristics

The implementation is conceptually straightforward but computationally expensive for high-order derivatives. A significant drawback is the difficulty in enforcing Neumann or Robin boundary conditions, which must be weakly imposed through the loss term, often leading to instability and poor convergence[31]. Despite its mesh-free elegance, the computational cost of AD scales poorly with the order of derivatives and the number of collocation points, making it prohibitive for complex, high-fidelity simulations.

2.2. The least-squares moving particle semi-implicit PINN (LSMPS-PINN)

The LSMPS-PINN framework integrates the high-order consistent spatial discretization schemes of the Least Squares Moving Particle Semi-implicit (LSMPS) method[7] into the physics-informed neural network paradigm. This approach replaces the computationally expensive AD for spatial derivatives with precomputed, sparse differentiation operators derived from a weighted least squares procedure, enhancing both efficiency and stability.

2.2.1. Mathematical framework

The core of the LSMPS method lies in its ability to construct arbitrary high-order consistent approximations of spatial derivatives using only a set of scattered nodal points (particles). This concept aligns with the general deterministic treatment proposed by Eldredge et al [32], who presented a unified approach for approximating spatial derivatives of any degree using integral operators on particle distributions. Consider a sufficiently smooth function $f(\mathbf{x})$ defined on a domain $\Omega \subset \mathbb{R}^d$. The goal is to approximate the derivatives of f at a point $\mathbf{x}_i$ based on its values at neighboring points $\mathbf{x}_j \in \Lambda_i$, where $\Lambda_i = \{j: \|\mathbf{x}_j - \mathbf{x}_i\| < r_e\}$ and r_e is a dilation parameter.

A local Taylor expansion of $f(\mathbf{x}_j)$ around $\mathbf{x}_i$ up to order p yields

$$f(\mathbf{x}_j) = f(\mathbf{x}_i) + \sum_{1 \leq |\alpha| \leq p} \left[\frac{1}{\alpha!} (\mathbf{x}_j - \mathbf{x}_i)^\alpha \partial^\alpha f(\mathbf{x}_i) \right] + R_{ij}^{p+1} \quad (5)$$

where α is a multi-index, ∂^α denotes the partial derivative corresponding to α , and R_{ij}^{p+1} is the residual term. Rearranging terms

$$f(\mathbf{x}_j) - f(\mathbf{x}_i) = \sum_{1 \leq |\alpha| \leq p} \left[\frac{1}{\alpha!} (\mathbf{x}_j - \mathbf{x}_i)^\alpha \partial^\alpha f(\mathbf{x}_i) \right] + R_{ij}^{p+1} \quad (6)$$

To avoid ill-conditioning in the least squares problem, a scaling parameter r_s (with $0 < r_s < r_e$) is introduced. Define the scaled relative coordinate $\mathbf{r}_{ji} = (\mathbf{x}_j - \mathbf{x}_i)/r_s$ and a polynomial basis vector $\mathbf{p}(\mathbf{r})$ containing monomials $\mathbf{r}^\alpha$ for $1 \leq |\alpha| \leq p$. The equation can be rewritten as

$$f(\mathbf{x}_j) - f(\mathbf{x}_i) = \mathbf{p}^T(\mathbf{r}_{ji}) [\mathbf{H}_s^{-1} d\mathbf{f}(\mathbf{x}_i)] + R_{ij}^{p+1} \quad (7)$$

where $d\mathbf{f}(\mathbf{x}_i)$ is a vector containing the derivatives $\partial^\alpha f(\mathbf{x}_i)$ for $1 \leq |\alpha| \leq p$, $\mathbf{H}_s = \text{diag}\{r_s^{-|\alpha|}\alpha!\}$ is a diagonal scaling matrix.

The weighted least squares method minimizes the sum of squared residuals $\sum_{j \in \Lambda_i} \left(R_{ij}^{p+1} \right)^2$, where $\omega_{ij} = \omega(\|\mathbf{x}_j - \mathbf{x}_i\|/r_e)$ is a weight function with compact support. This leads to the normal equations

$$\mathbf{M}_i [\mathbf{H}_s^{-1} d\mathbf{f}(\mathbf{x}_i)] = \mathbf{b}_i \quad (8)$$

$$\mathbf{b}_i = \sum_{j \in \Lambda_i} \omega_{ij} \mathbf{p}(\mathbf{r}_{ji}) [f(\mathbf{x}_j) - f(\mathbf{x}_i)] \quad (9)$$

$$\mathbf{M}_i = \sum_{j \in \Lambda_i} \omega_{ij} \mathbf{p}(\mathbf{r}_{ji}) \mathbf{p}^T(\mathbf{r}_{ji}) \quad (10)$$

where $\mathbf{b}_i$ is the right-hand side vector, $\mathbf{M}_i$ is the moment matrix.

Solving for the derivatives

$$d\mathbf{f}^h(\mathbf{x}_i) = \mathbf{H}_s [\mathbf{M}_i^{-1} \mathbf{b}_i] \quad (11)$$

which defines the standard LSMPS scheme (Type-A) for spatial derivatives[20]. For instance, the gradient and Laplacian can be extracted from the vector $d\mathbf{f}^h(\mathbf{x}_i)$ by selecting the appropriate components corresponding to first-order and second-order derivatives, respectively.

2.2.2. LSMPS-PINN formulation

In LSMPS-PINN, a neural network $u_0(\mathbf{x}, t)$ is trained to approximate the solution of a PDE. However, instead of using AD to compute spatial derivatives in the PDE residual, the LSMPS discrete operators are

employed. The key differential operators are constructed as follows[15, 19, 20]

$$\langle \nabla \phi \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\phi_j - \phi_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (12)$$

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j - \mathbf{u}_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (13)$$

$$\langle \nabla^2 \phi \rangle_i = \sum_{j \neq i} \left\{ 2 \frac{w_{ij}}{r_s^2} (\phi_j - \phi_i) [\mathbf{L}_3 + \mathbf{L}_4] \mathbf{p}_{ij}^T \right\} \quad (14)$$

where $\mathbf{L}_k$ represents the k -th row of the matrix $\mathbf{M}_i^{-1}$. For a general time-dependent PDE

$$\frac{\partial u}{\partial t} = L[u] \quad (15)$$

where L is a spatial differential operator, the physics-informed loss function in LSMPS-PINN is constructed as

$$L_{\text{phys}}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \left\| \left(\frac{\partial u_0}{\partial t} \right)_i - L^h[u_0]_i \right\|^2 \quad (16)$$

Here, L^h denotes the discrete approximation of L using the LSMPS operators shown above. For example, if $L[u] = \nu \nabla^2 u$ (diffusion term), then the discrete operator becomes

$$L^h[u_0]_i = \nu \sum_{j \neq i} \left\{ 2 \frac{w_{ij}}{r_s^2} (u_{0,j} - u_{0,i}) [\mathbf{L}_3 + \mathbf{L}_4] \mathbf{p}_{ij}^T \right\} \quad (17)$$

This approach eliminates the need for automatic differentiation in spatial derivative computations, significantly reducing computational overhead while maintaining mathematical consistency.

2.2.3. Implementation and advantages

The LSMPS-PINN implementation involves particle-based domain discretization with characteristic spacing h , followed by precomputation of the inverse moment matrices $\mathbf{M}_i^{-1}$ for each particle neighborhood. The differential operators are assembled as sparse matrices prior to network training, enabling efficient computation during the optimization process. The neural network training minimizes a combined loss function that incorporates physical residuals along with boundary and initial conditions.

The key advantages of LSMPS-PINN include its ability to achieve arbitrary high-order accuracy through polynomial order selection, robust handling of boundary conditions via constrained least squares formulations, and significantly improved computational efficiency through precomputed sparse operators. Unlike AD-PINN that suffers from recursive graph traversal costs for high-order derivatives, LSMPS-PINN provides $O(1)$ complexity per training step after the initial pre-computation phase, while the mathematically consistent discretization schemes enhance numerical stability and convergence behavior. This methodology demonstrates how traditional numerical analysis techniques can be effectively integrated with modern deep learning approaches to overcome fundamental limitations in scientific machine learning.

2.3. The least-squares finite-volume particle PINN (LSFVP-PINN)

While the LSMPS-PINN framework described above offers significant computational advantages, its reliance on strong-form collocation requires the direct approximation of second-order derivatives (e.g., the Laplacian) on scattered points. High-order derivative approximations are inherently sensitive to particle distribution irregularities, which can limit numerical precision in complex flow regimes.

To address this limitation, we introduce the LSFVP-PINN. This novel hybrid framework synergistically integrates the integral formulation of

the finite volume method with the high-order reconstruction accuracy of meshless particle approaches. By coupling particle-style local reconstructions with finite-volume-style flux assembly, LSFVP-PINN transforms the target PDEs into their integral forms. This allows the application of the divergence theorem to reduce the order of the required spatial derivatives (e.g., from Laplacian to gradient fluxes), thereby improving the robustness and accuracy of the approximation.

2.3.1. Mathematical formulation

The core innovation of LSFVP-PINN lies in its use of the Type-B LSMPS scheme for face-centered flux reconstruction within a finite volume framework. Consider a computational domain Ω discretized by a fixed set of particles $\{\mathbf{x}_i\}$ with characteristic spacing h . For each particle, a control volume V_i is defined, and the governing PDE is integrated over V_i

$$\frac{\partial}{\partial t} \int_{V_i} u dV + \oint_{\partial V_i} \mathbf{F}(u) \cdot \mathbf{n} dS = 0 \quad (18)$$

where $\mathbf{F}(u)$ is the flux function. Applying the divergence theorem yields the semi-discrete form

$$\frac{du_i}{dt} |V_i| + \sum_{f \in \partial V_i} \mathbf{F}_f \cdot \mathbf{n}_f S_f = 0 \quad (19)$$

where the summation is over the faces of the control volume. The flux $\mathbf{F}_f$ at each face center is reconstructed using the Type-B LSMPS scheme, which provides high-order approximations of derivatives based on neighboring particle values.

The Type-B LSMPS scheme derives from a weighted least squares procedure. For a function $\Phi(\mathbf{x})$, its derivatives at a point $\mathbf{x}_i$ are approximated using values at neighboring points $\mathbf{x}_j \in \Lambda_i = \{j: \|\mathbf{x}_j - \mathbf{x}_i\| < r_e\}$. The Taylor expansion up to order p around $\mathbf{x}_i$ is:

$$\Phi(\mathbf{x}_j) = \Phi(\mathbf{x}_i) + \sum_{1 \leq |\alpha| \leq p} \left[\frac{1}{\alpha!} (\mathbf{x}_j - \mathbf{x}_i)^\alpha D^\alpha \Phi(\mathbf{x}_i) \right] + R_{ij}^{p+1} \quad (20)$$

where α is a multi-index. The Type-B formulation introduces a scaled coordinate system to improve conditioning

$$\mathbf{r} = \frac{\mathbf{x}_j - \mathbf{x}_i}{r_s} \quad (21)$$

where r_s is a scaling parameter. The basis vector $\mathbf{p}(\mathbf{r})$ for second-order completeness ($p=2$) in two dimensions is

$$\tilde{\mathbf{p}}_{ij} = \tilde{\mathbf{p}}\left(\frac{\mathbf{r}_{ij}}{r_s}\right) = \left[1, \frac{x_{ij}}{r_s}, \frac{y_{ij}}{r_s}, \frac{x_{ij}^2}{r_s^2}, \frac{y_{ij}^2}{r_s^2}, \frac{x_{ij}y_{ij}}{r_s^2} \right] \quad (22)$$

The moment matrix $\mathbf{M}_i$ is constructed through a weighted summation over neighboring particles within an influence radius r_e

$$\mathbf{M}_i = \sum_{j \in \Lambda_i} \omega(\|\mathbf{x}_j - \mathbf{x}_i\|) \mathbf{p}\left(\frac{\mathbf{x}_j - \mathbf{x}_i}{r_s}\right) \mathbf{p}^T\left(\frac{\mathbf{x}_j - \mathbf{x}_i}{r_s}\right) \quad (23)$$

where $\omega(r) = (1 - r/r_e)^2$ for $r < r_e$ is the weighting function. The derivative operators are then obtained through

$$D^\alpha \Phi(\mathbf{x}_i) \approx \mathbf{H}_i \mathbf{M}_i^{-1} \sum_{j \in \Lambda_i} \omega(\|\mathbf{x}_j - \mathbf{x}_i\|) \mathbf{p}\left(\frac{\mathbf{x}_j - \mathbf{x}_i}{r_s}\right) \Phi(\mathbf{x}_j) \quad (24)$$

$$\tilde{\mathbf{H}}_i = \text{diag}\left(1, \frac{1}{r_s}, \frac{1}{r_s}, \frac{2}{r_s^2}, \frac{2}{r_s^2}, \frac{1}{r_s^2}\right) \quad (25)$$

2.3.2. Finite volume formulation with type-b operators

The LSFVP-PINN methodology leverages the Type-B operators within a finite volume framework to achieve both high-order accuracy and robust flux-based discretization. The key differential operators for

gradient, divergence, and Laplacian computations are constructed as follows

$$\langle \nabla \phi \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\phi_j - \phi_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (26)$$

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \sum_{j \neq i} \left\{ \frac{w_{ij}}{r_s} (\mathbf{u}_j - \mathbf{u}_i) \left(\begin{bmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \end{bmatrix} \mathbf{p}_{ij}^T \right) \right\} \quad (27)$$

$$\langle \nabla^2 \phi \rangle_i = \sum_{j \neq i} \left\{ 2 \frac{w_{ij}}{r_s^2} (\phi_j - \phi_i) [\mathbf{L}_3 + \mathbf{L}_4] \mathbf{p}_{ij}^T \right\} \quad (28)$$

where L_k denotes the k -th row of the matrix $\hat{\mathbf{M}}_i^{-1}$. Within the physics-informed neural network framework, a neural network $u_0(\mathbf{x}, t)$ parameterized by θ is trained to approximate the solution of the governing PDE. The physics-informed loss function incorporates the finite volume discretization

$$L_{phys}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \left\| \frac{du_{\theta,i}}{dt} + \frac{1}{|V_i|} \sum_{f \in \partial V_i} \mathbf{F}_f \cdot \mathbf{n}_f S_f \right\|^2 \quad (29)$$

where the flux term $\mathbf{F}_f$ is reconstructed using the Type-B LSMPS operators at face centers. Unlike strong-form collocation, this integral approach aggregates flux information across control volume boundaries. While strict discrete flux cancellation is not fully enforced due to the meshless topology, the reduction in derivative order—calculating first-order fluxes instead of second-order operators—yields a significantly more stable and accurate numerical scheme.

2.3.3. Implementation and comparative advantages

The implementation of LSFVP-PINN follows a two-phase approach: a precomputation phase where sparse Type-B operators are assembled, and a training phase where these operators efficiently compute residuals. This method exhibits superior numerical stability, particularly for convection-dominated flows, and offers more accurate enforcement of boundary conditions via face-centered flux reconstruction. While computationally more intensive per step than LSMPS due to the flux assembly, LSFVP provides the highest physical fidelity among the compared methods.

3. Numerical tests

Unlike traditional grid-based methods (e.g., FVM) where uniform h -refinement strictly guarantees monotonic error reduction, simply increasing the collocation point density in PINNs often complicates the highly non-convex loss landscape. As highlighted in recent literature[9], this exacerbates gradient pathologies and optimization difficulties, which can hinder stable asymptotic convergence. Consequently, all subsequent cases are evaluated at a fixed, representative resolution to fairly compare the relative performance of AD-PINN, LSMPS-PINN, and LSFVP-PINN under identical optimization conditions.

3.1. Two-dimensional viscous burgers equation

We first validate the proposed LSFVP-PINN framework on a manufactured two-dimensional viscous Burgers problem. The governing scalar equation is

$$u_t + \nabla \cdot \left(\frac{1}{2} u^2 \right) - \nu \Delta u = S(\mathbf{x}, t) \quad (30)$$

where $\mathbf{x} = (x, y) \in [0, 1]^2, t \in (0, T]$ and the forcing term $S(\mathbf{x}, t)$ is chosen so that the analytical solution is

$$u_{\text{exact}}(x, y, t) = e^{-2\pi^2 t} \sin(\pi x) \cos(\pi y) \quad (31)$$

Dirichlet boundary conditions are imposed on the whole boundary

from the analytic solution, and the initial state $isu(\mathbf{x}, 0) = u_{exact}(\mathbf{x}, 0)$. In all experiments reported below the kinematic viscosity is $\nu = 0.01$ and the final evaluation time is $T = 0.01$.

A common network architecture and training recipe were used across all comparative methods to ensure a fair assessment: a fully connected feed-forward network with four hidden layers and 60 neurons per layer, SiLU activations, and the Adam optimizer with learning rate 10^{-3} . The spatial domain is discretized with a regular collocation of 129×129 points ($n=16,641$, $h=1/128$). For the meshless operator-based methods (LSMPS-PINN and LSFVP-PINN), the influence radius is set to $r_e = 3.1h$ and the scaling factor is set to $r_s = 0.2h$. These parameter settings are selected based on the stability analysis and optimal recommendations provided by Tamai and Koshizuka[20], ensuring a well-conditioned moment matrix while maintaining sufficient local connectivity. The transient experiments use a timestep chosen as $\Delta t = T/[T/h^2]$ so that Δt is $O(h^2)$. Training ran for 20,000 epochs for each tested method; boundary conditions were enforced strongly by replacing boundary labels with the analytic values or by a boundary penalty in the loss. The three methods compared are (i) AD-PINN, the classical residual-based PINN using automatic differentiation to evaluate all spatial derivatives; (ii) LSMPS-PINN, a center-based meshless least-squares operator embedded as numerical labels; and (iii) LSFVP-PINN, the proposed face-reconstruction + flux-assembly operator used to generate integral-based numerical labels and to perform implicit diffusion corrections in a predictor–corrector time stepping.

Quantitative performance is reported in Table 1. We measure the prediction error at T by the root-mean-square (RMS) norm

$$\frac{\|u_{pred} - u_{exact}\|_2}{\sqrt{n}} \quad (32)$$

and record training time together with the main computational component times (autograd accumulation for AD-PINN; sparse-operator work for operator-based methods). The results show that LSFVP-PINN attains the best accuracy and that both operator-based methods substantially reduce the autograd burden and overall time compared with AD-PINN.

The quantitative performance metrics in Table 1 delineate a clear efficiency–accuracy trade-off between the proposed frameworks. LSMPS-PINN exhibits the highest computational throughput, completing the training in 516.72 s with the lowest sparse operator overhead ($t_{sparse} = 282.45$ s). This makes it the most efficient choice for scenarios where rapid turnaround is prioritized. Conversely, LSFVP-PINN trades a portion of this computational speed for enhanced physical fidelity, achieving the lowest RMS error. This superior accuracy is attributed to its integral formulation, which bypasses the direct calculation of second-order derivatives required by LSMPS-PINN. By reconstructing lower-order fluxes at face centers, LSFVP-PINN mitigates the discretization errors associated with high-order operators on irregular point clouds. Compared to the baseline AD-PINN, both operator-based methods offer substantial improvements: LSMPS-PINN reduces the total training time by approximately 67 %, while LSFVP-PINN reduces the error by nearly an order of magnitude. This result confirms that the two proposed methods occupy distinct niches: LSMPS for speed and LSFVP for precision.

The superior performance of LSFVP-PINN is further elucidated through diagnostic analysis of the centerline profiles and training dynamics. Fig. 1 presents centerline velocity distributions along the

domain diagonal ($x = y$), comparing numerical solutions against the analytical benchmark. The LSFVP-PINN solution exhibits excellent agreement with the exact profile, particularly in high-gradient regions where it maintains superior phase accuracy and amplitude preservation relative to the baseline AD-PINN, while offering sharper gradient definition than the smoother LSMPS-PINN profile. This enhanced accuracy stems from the face-based flux reconstruction scheme, which provides improved conservation properties.

The training convergence characteristics shown in Fig. 2 provide additional insights into method efficiency. LSFVP-PINN demonstrates rapid and stable convergence, reaching a loss plateau of approximately 10^{-4} within 20,000 epochs. The consistent downward trajectory without significant oscillations indicates robust optimization behavior, attributed to the balanced loss components and effective gradient propagation through the operator-based approach.

To address the practical concern regarding the solver's performance under higher Reynolds numbers, we conducted a sensitivity analysis by reducing the kinematic viscosity to $\nu = 0.001$. In this regime, the Burgers equation becomes convection-dominated, leading to the development of sharper gradients that pose significant challenges for neural network-based solvers due to the spectral bias phenomenon[13]. The experiment was conducted on the same domain with a prolonged simulation time $T=0.5$ to allow the convection effects to fully evolve. The exact solution is prescribed as:

$$u(x, y, t) = e^{-t} \sin(\pi x) \sin(\pi y) \quad (33)$$

The comparative results are summarized in Table 2

As expected, the reduced viscosity leads to a general increase in prediction errors for the baseline method. The AD-PINN exhibits a noticeable degradation in accuracy. However, the proposed operator-based frameworks demonstrate remarkable robustness. LSFVP-PINN achieves the lowest error. This superior accuracy is physically justified by the integral formulation of LSFVP. LSMPS-PINN continues to lead in computational efficiency, finishing the training in just 233 s—nearly 4 times faster than AD-PINN—while maintaining an acceptable error level.

Fig. 3 presents the centerline velocity profiles for this convection-dominated case. The AD-PINN solution fails to capture the sharp peak of the wave, exhibiting excessive numerical dissipation. In contrast, the LSFVP-PINN solution preserves the wave amplitude and phase accuracy, confirming that the flux-based operator learning strategy is particularly advantageous for problems with dominant convective transport.

In summary, the combination of quantitative metrics and diagnostic visualizations confirms that the proposed operator-based frameworks offer flexible solutions for different computational priorities. While LSMPS-PINN demonstrates exceptional efficiency for rapid flow prediction, LSFVP-PINN ensures maximum physical fidelity for conservation-critical regimes. Collectively, these formulations successfully address key limitations of both residual-based and conventional meshless approaches, establishing a versatile framework for complex flow simulations.

3.2. Two-dimensional diffusion equation

3.2.1. Solving the diffusion equation with Dirichlet boundary conditions

We validate the proposed LSFVP-PINN framework on a manufactured two-dimensional diffusion problem. The governing equation is

$$\frac{\partial u}{\partial t} - \nu \Delta u = S(\mathbf{x}, t) \quad (34)$$

where $\mathbf{x} = (x, y) \in [0, 1]^2$, $t \in (0, T]$, and the forcing term $S(\mathbf{x}, t)$ is chosen so that the analytical solution is

$$u_{exact}(\mathbf{x}, y, t) = e^{-2\pi^2 \nu t} \sin(\pi x) \sin(\pi y) \quad (35)$$

Dirichlet boundary conditions are imposed on the whole boundary

Table 1

Burgers test — training time breakdown and final error.

Method	Total training time(s)	Autograd / sparse time (s)	RMS error
AD-PINN	1559.60	autograd = 931.56	4.41×10^{-2}
LSMPS-PINN	516.72	sparse = 282.45	6.97×10^{-3}
LSFVP-PINN	657.59	sparse = 222.98	4.66×10^{-3}

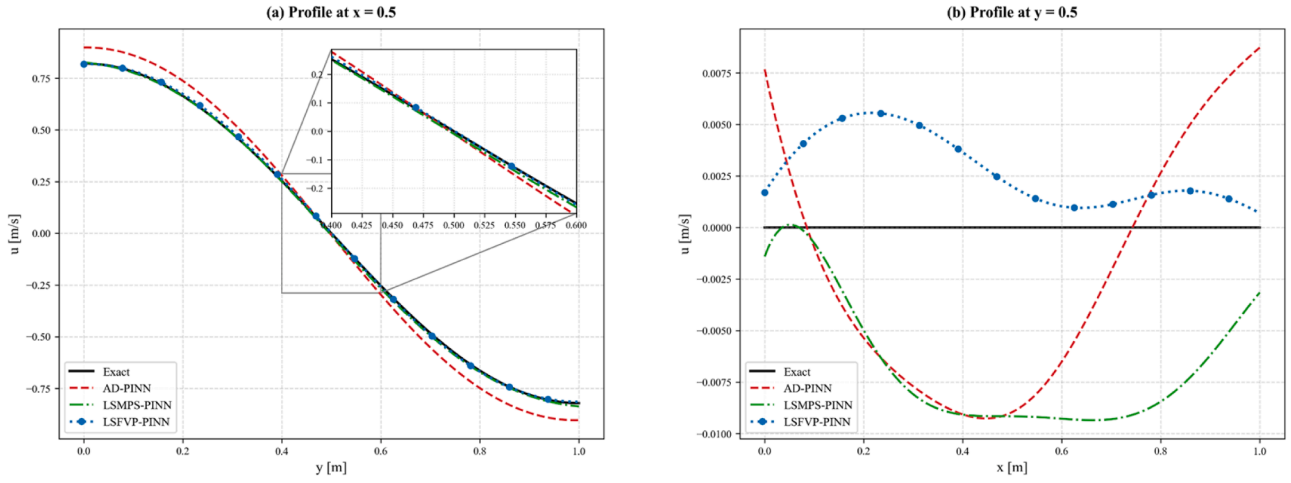
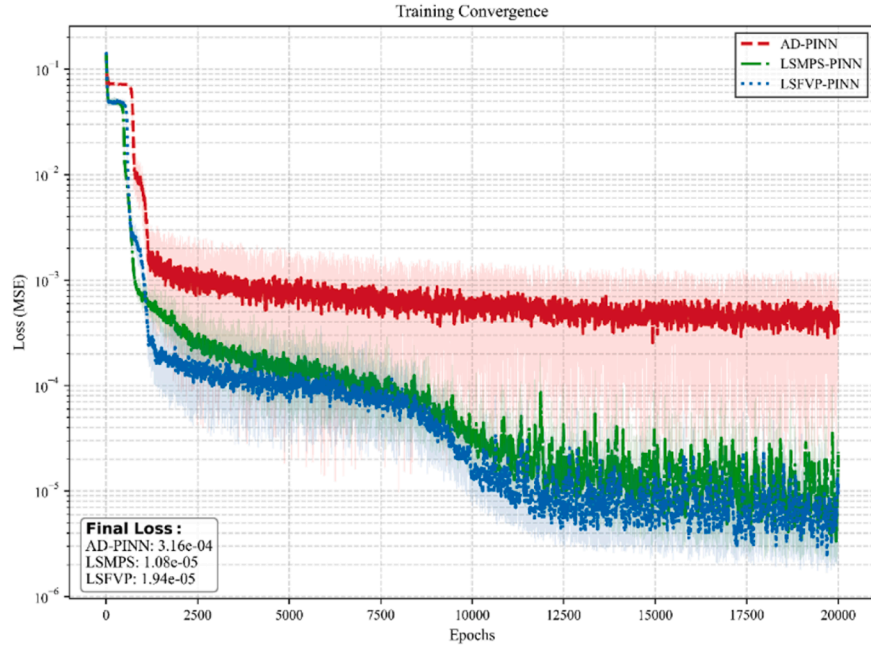
Fig. 1. Centerline velocity profiles at $T = 0.01$.

Fig. 2. Training loss convergence history of burgers test.

Table 2

Performance comparison for the Burgers equation with reduced viscosity.

Method	Total training time(s)	Autograd / sparse time (s)	RMS error
AD-PINN	852.10	autograd = 314.48	1.39×10^{-2}
LSMPS-PINN	233.82	sparse = 18.04	6.92×10^{-3}
LSFVP-PINN	278.81	sparse = 21.75	5.05×10^{-3}

from the analytic solution, and the initial state $isu(x, 0) = u_{exact}(x, 0)$. In all experiments reported below the diffusion coefficient is $\nu = 1.0$ and the final evaluation time is $T = 0.01$.

The same network architecture and training parameters from Section 3.1 are employed: a four-hidden-layer fully connected network with 60 neurons per layer, SiLU activations, Adam optimizer with learning rate 10^{-3} , and strong enforcement of boundary conditions. Spatial discretization uses a regular 129×129 grid ($n=16,641$, $h=1/128$), with timestep $\Delta t = 1.524 \times 10^{-5}$ selected to maintain numerical stability. Training proceeds for 10,000 epochs for each method.

We compare the same three approaches: (i) AD-PINN using

automatic differentiation for derivative computations; (ii) LSMPS-PINN with center-based meshless operators; and (iii) LSFVP-PINN employing our face-reconstruction and flux-assembly formulation.

The performance breakdown reveals distinct computational characteristics tailored to different modeling needs. As shown in Table 3, LSMPS-PINN demonstrates superior training efficiency, achieving a convergent solution in just 146.30 s—approximately 39 % faster than LSFVP-PINN and 3.5 times faster than AD-PINN. This highlights the efficiency of the center-based strong-form formulation for smooth diffusive problems. On the other hand, LSFVP-PINN delivers the highest numerical accuracy (RMS error 5.92×10^{-3}), validating the benefit of its finite-volume-based conservation properties even in linear regimes. The convergence behaviors reinforce this complementarity: LSMPS provides a rapid approximation, while LSFVP utilizes the integral form to reduce derivative sensitivity, yielding a lower error floor at the cost of increased operator complexity.

The training convergence behavior shows similar patterns to the Burgers equation case, with LSFVP-PINN maintaining stable convergence throughout the training process. The method's operator-based

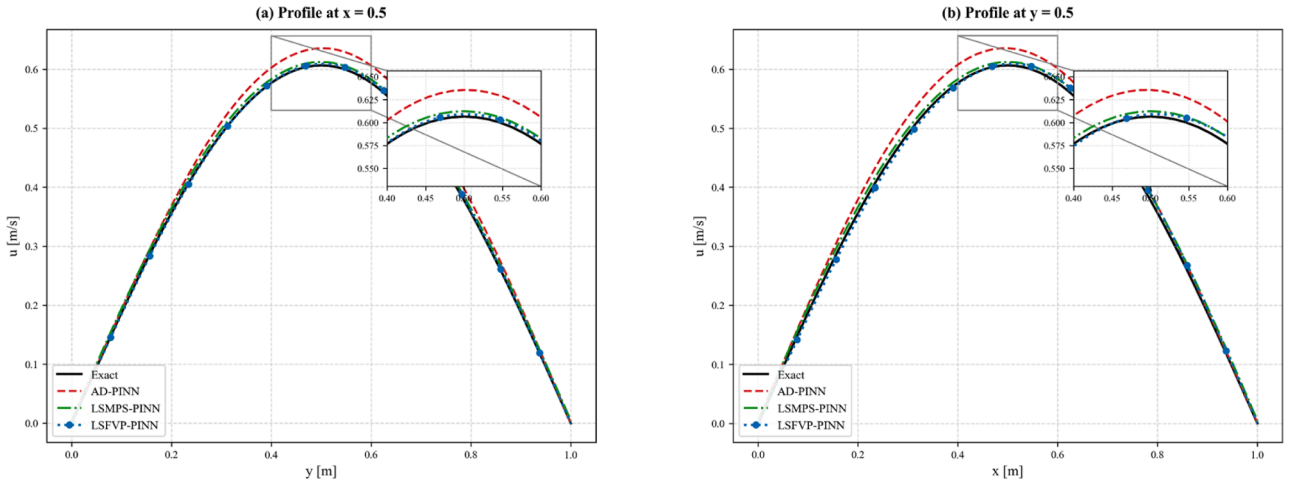


Fig. 3. Centerline velocity profiles for the convection-dominated Burgers equation at $T=0.5$.

Table 3

Diffusion test — training time breakdown and final error.

Method	Total training time(s)	Autograd / sparse time (s)	RMS error
AD-PINN	511.33	autograd = 269.41	5.27×10^{-1}
LSMPS-PINN	146.30	sparse = 8.79	1.34×10^{-2}
LSFVP-PINN	239.87	sparse = 76.11	5.92×10^{-3}

formulation effectively balances computational efficiency with numerical accuracy, demonstrating particular advantage for diffusion-dominated problems where derivative computations constitute the primary computational burden.

The quantitative results presented in Table 3 are further elucidated through detailed visual comparisons of the solution profiles and training dynamics.

Fig. 4 displays the centerline profiles of the scalar field along $x = 0.5$ and $y = 0.5$ at the final time $T = 0.0100$, comparing the predictions of the three PINN methodologies against the exact analytic solution. A clear hierarchy in accuracy is observed, corroborating the RMS error metrics from Table 3. The AD-PINN prediction exhibits significant deviation from the exact solution, particularly near the peaks and troughs of the sinusoidal profile. This substantial error is consistent with its high final L_2 error of 0.527, indicating a fundamental difficulty in accurately capturing the diffusion physics via automatic differentiation for this problem. In contrast, both the LSMPS-PINN and LSFVP-PINN predictions align remarkably well with the exact solution across the entire domain. The LSFVP-PINN result shows almost perfect superposition with the

analytic curve, visually confirming its superior accuracy and its position as the most accurate method for this test case.

The training behavior and convergence characteristics of the different methods are critically examined in Fig. 5, which plots the loss history versus training epochs on a logarithmic scale. The AD-PINN curve demonstrates poor convergence, with the loss oscillating around 10^{-2} throughout the training process and never stabilizing to a lower value. This instability and failure to converge effectively explain the poor predictive accuracy observed in Fig. 4 and Table 3. Conversely, the operator-based methods, LSMPS-PINN and LSFVP-PINN, exhibit robust, stable, and rapid convergence. Both curves descend smoothly over several orders of magnitude, plateauing at an exceptionally low loss level of approximately 10^{-6} . This stark contrast underscores a key advantage of the operator-based approach: by providing numerically consistent and physically meaningful labels, they create a more stable and well-conditioned optimization landscape for the neural network, leading to successful training and high accuracy.

3.2.2. Solving the diffusion equation with mixed boundary conditions

To further evaluate the robustness of the proposed frameworks in scenarios with more complex boundary conditions, we consider the diffusion equation on the unit square $\Omega = [0, 1]^2$ with mixed Dirichlet and Neumann boundary conditions. The governing equation remains the same as in Eq. (24), but the boundary conditions are defined as follows:

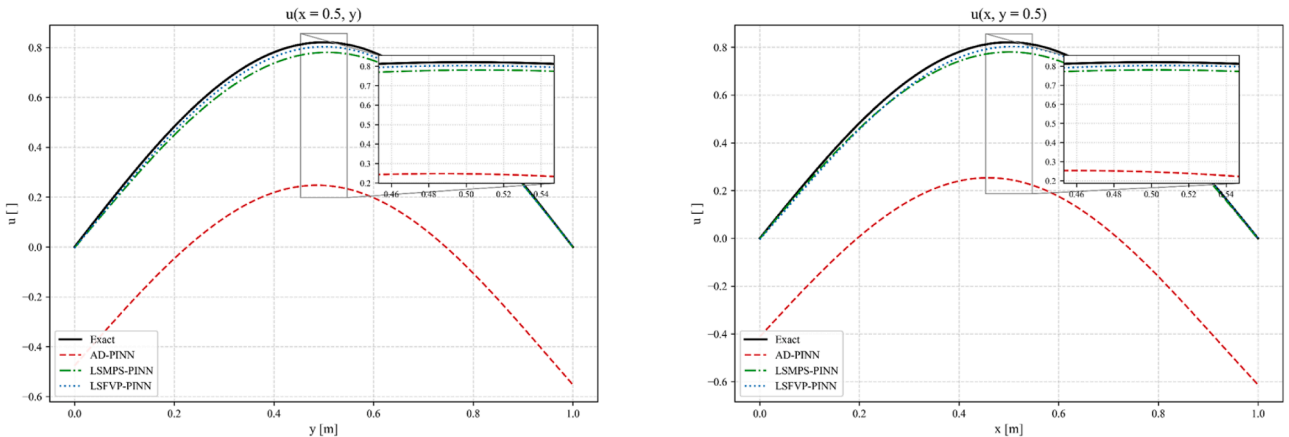


Fig. 4. Centerline velocity profiles at $T = 0.01$.

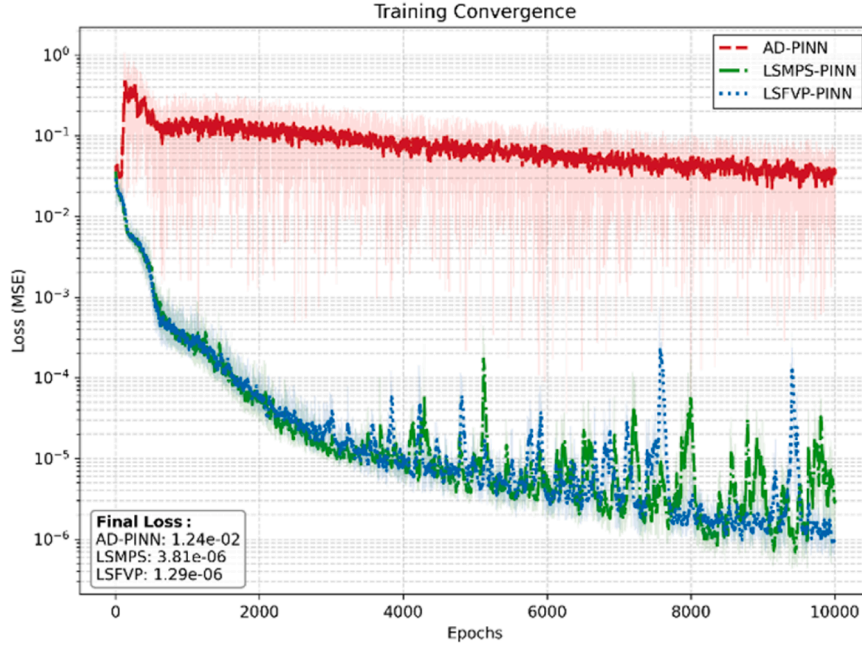


Fig. 5. Training loss convergence history of Diffusion test.

$$\begin{cases} u(0, y, t) = u(x, 0, t) = u(x, 1, t) = 0 \\ \frac{\partial u}{\partial x}(1, y, t) = 0 \end{cases} \quad (36)$$

The analytical solution is constructed as $u(x, y, t) = e^{-1.25\pi^2 \nu t} \sin(0.5\pi x) \sin(\pi y)$, which satisfies the homogeneous Neumann condition at the right boundary ($x = 1$). The kinematic viscosity is set to $\nu = 1.0$, and the simulation runs to $T = 0.01$. The discretization parameters remain consistent with the previous cases.

Table 4 summarizes the computational performance and accuracy metrics. A significant performance divergence is observed compared to the pure Dirichlet case. The standard AD-PINN struggles to converge effectively, yielding a high relative L_2 error of 3.32×10^{-1} . In contrast, both operator-based frameworks maintain robustness. LSFVP-PINN achieves the best performance in terms of both accuracy and computational efficiency. LSMPS-PINN also performs reliably with an error of 6.01×10^{-2} , demonstrating that strong-form collocation remains a viable strategy for mixed boundary problems, though slightly less accurate than the integral-based formulation of LSFVP in this specific regime.

Fig. 6 illustrates the centerline velocity profiles. The advantage of the proposed methods is visually evident: AD-PINN fails to capture the peak amplitude of the scalar field, exhibiting significant diffusive error. Conversely, both LSFVP-PINN and LSMPS-PINN profiles closely track the analytical solution. The zoom-in views confirm that the operator-based methods correctly resolve the zero-gradient condition at the right boundary and the value constraint at the boundaries, validating their superior applicability to problems involving mixed boundary conditions.

Table 4

Diffusion test (Dirichlet and Neumann boundary conditions)— training time breakdown and final error.

Method	Total training time(s)	Autograd / sparse time (s)	RMS error
AD-PINN	2197.32	autograd = 1329.71	3.32×10^{-1}
LSMPS-PINN	899.59	sparse = 35.84	6.01×10^{-2}
LSFVP-PINN	540.97	sparse = 34.54	5.51×10^{-2}

3.2.3. Validating the methods on irregular point distributions

The results presented in Section 3.2 were obtained using regular Cartesian layouts to establish a controlled baseline. However, a primary theoretical advantage of meshless particle methods is their geometric flexibility, allowing them to operate on unstructured or irregular point distributions. To empirically validate this capability for the proposed PINN frameworks, we conducted an additional sensitivity test on the diffusion equation.

We introduced a random perturbation to the interior collocation points while keeping the boundary nodes fixed to preserve the domain geometry. The coordinates of the internal particles were displaced as:

$$x'_i = x_i + \delta_x, y'_i = y_i + \delta_y \quad (37)$$

where δ_x, δ_y are random variables drawn from a uniform distribution within $[-0.2h, 0.2h]$, representing a 20 % irregularity in the particle arrangement. Table 5 compares the performance of the proposed methods on this irregular point cloud against the regular grid benchmark.

As Table 5 showed, both operator-based methods demonstrate remarkable robustness. The LSMPS-PINN error increases only marginally from 1.34×10^{-2} to 2.07×10^{-2} , proving that the strong-form least-squares operators effectively adapt to local topological distortions. Similarly, LSFVP-PINN maintains its superior accuracy range, confirming that the flux reconstruction scheme remains stable even when control volumes become slightly distorted. These results provide concrete numerical evidence supporting the claim that the proposed frameworks inherit the geometric flexibility inherent to meshless particle methods.

3.2.4. Comparison with traditional finite volume method

To comprehensively evaluate the performance of the proposed frameworks against established numerical techniques, we conducted a comparative study with a traditional Finite Volume Method (FVM) solver. The FVM solver employs a standard five-point stencil for spatial discretization and the Crank-Nicolson scheme for time integration on the same 129×129 regular grid used in Section 3.2.1. The quantitative results are summarized in Table 6.

As expected, the FVM solver achieves high numerical precision with an L_2 error of approximately 10^{-6} . In the current experimental setup, the absolute accuracy of the PINN-based frameworks has not yet reached

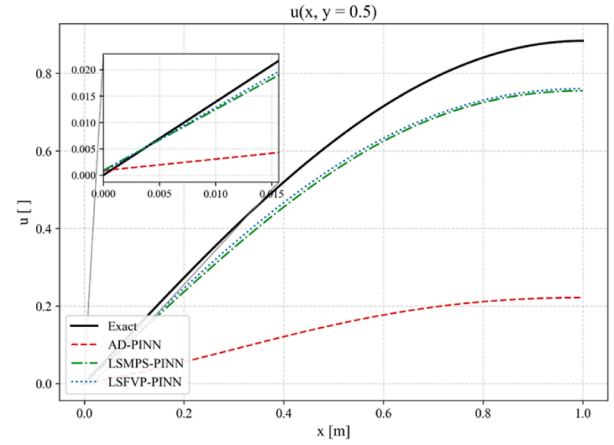
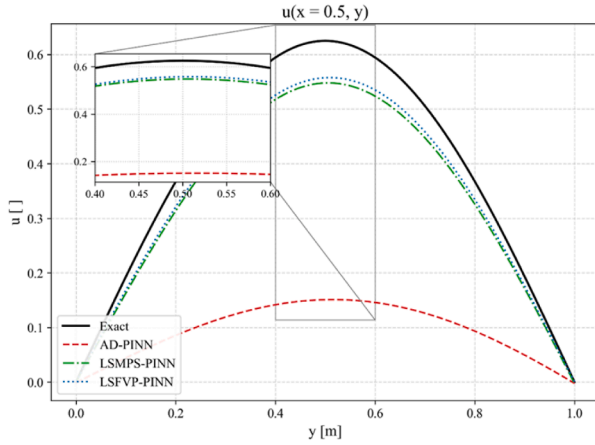


Fig. 6. Centerline profiles of the solution u at $T=0.01$ for the diffusion problem with mixed boundary conditions.

Table 5
Comparison of L_2 errors on regular vs. irregular grids.

Method	Grid Topology	Relative L_2 Error
LSMPS-PINN	Regular	1.34×10^{-2}
LSMPS-PINN	Irregular	2.07×10^{-2}
LSFVP-PINN	Regular	5.92×10^{-3}
LSFVP-PINN	Irregular	1.11×10^{-2}

Table 6
Comparison between Traditional FVM and PINN frameworks.

Method	Inference Time(s)	L_2 Error
FVM	2.01	4.03×10^{-6}
AD-PINN	2.70×10^{-3}	5.27×10^{-1}
LSMPS-PINN	2.50×10^{-3}	1.34×10^{-2}
LSFVP-PINN	2.80×10^{-3}	5.92×10^{-3}

this ultimate precision of the traditional method. This discrepancy is primarily attributed to the non-convex nature of neural network optimization, the spectral bias phenomenon inherent to deep learning[13], and the strictly constrained training budget, such as fixed learning rates and limited training epochs. The accuracy of these neural solvers is not fundamentally capped and can be progressively improved by extending the training iterations, employing second-order optimizers (e.g., L-BFGS), or increasing the network capacity. This represents a speedup of over 700 times compared to the iterative FVM solver. Such a characteristic is highly valuable for engineering applications requiring real-time control, uncertainty quantification, or inverse problem solving where repeated rapid queries are necessary.

Despite the initial gap in absolute accuracy, the proposed operator-based PINNs demonstrate a distinct and overwhelming advantage in terms of online inference efficiency. The traditional FVM requires serial time-stepping from the initial state to reach the target time T , taking approximately 2.01 s for a single run. In contrast, once the offline training is completed, the LSFVP-PINN acts as a fast surrogate model capable of predicting the instantaneous state at any query time t in approximately 2.80×10^{-3} s.

Furthermore, it must be emphasized that PINN frameworks and traditional numerical methods are not mutually exclusive or substitutive; rather, they share a fundamentally complementary relationship[5, 7]. Traditional methods like FVM excel in delivering ultra-high precision for forward modeling on fixed discrete grids. Conversely, neural PDE solvers exhibit unique value in scenarios where traditional methods encounter bottlenecks, such as acting as ultra-fast surrogate models, assimilating noisy data, and handling complex geometries without the need for cumbersome mesh generation[5].

In conclusion, the diffusion equation case study reinforces the findings from the Burgers equation analysis. It demonstrates that the proposed framework provides robust alternatives to AD-PINN: LSMPS-PINN excels in minimizing computational cost, making it suitable for rapid prototyping, while LSFVP-PINN prioritizes numerical precision, showing particular utility in applications where error minimization is paramount. These distinct characteristics allow researchers to select the optimal solver based on the specific accuracy-efficiency constraints of the problem.

3.3. Lid-driven cavity flow at $Re = 100$

The lid-driven cavity flow represents a classic benchmark in computational fluid dynamics, testing the ability of numerical methods to handle incompressible flow with strong boundary interactions. We consider the steady-state, two-dimensional Navier-Stokes equations for an incompressible fluid

$$\mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \Delta \mathbf{u} \quad (38)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (39)$$

where $\mathbf{u} = (u, v)$ denotes the velocity field, represents pressure, and $Re = 100$ is the Reynolds number. The problem is solved on a unit square domain Ω . The boundary conditions enforce a no-slip condition ($u = v = 0$) along the bottom and vertical walls. The top boundary ($y = 1$) drives the flow with a prescribed tangential velocity $u = 1$ and vanishing normal velocity $v = 0$.

The performance metrics summarized in Table 7 demonstrate that both operator-based frameworks successfully mitigate the computational bottleneck of AD-PINN, reducing the total training time by approximately 65–67 %. The comparison between LSMPS-PINN and LSFVP-PINN further illustrates their complementary nature. LSMPS-PINN maintains its advantage in training speed (922.74 s) while achieving an accuracy level (6.74×10^{-4}) that is highly competitive with LSFVP-PINN (6.27×10^{-4}). While LSFVP-PINN marginally outperforms LSMPS-PINN in capturing the sharp gradients near the singular corners (as expected from its integral-based formulation which reduces the required derivative order), LSMPS-PINN offers a more lightweight

Table 7
Lid-Driven cavity test — training time breakdown and final error.

Method	Total training time(s)	MSE vs Ghia Benchmark
AD-PINN	2830.32	7.15×10^{-4}
LSMPS-PINN	922.74	6.74×10^{-4}
LSFVP-PINN	974.71	6.27×10^{-4}

alternative that captures the global flow topology with nearly identical fidelity but lower computational cost.

Fig. 7 presents a qualitative comparison of the streamline topologies obtained using the standard AD-PINN and the proposed Type-B operator-based frameworks (LSMPS-PINN and LSFVP-PINN). As observed, the flow structures predicted by the three methods exhibit a high degree of topological consistency. All models accurately capture the characteristic features of the steady-state solution at $Re = 100$. Notably, the LSFVP-PINN and LSMPS-PINN methods, which rely on discrete meshless operators for spatial derivatives, produce streamlines that are visually indistinguishable from those generated by the AD-PINN using exact automatic differentiation. This indicates that the proposed Type-B operators effectively handle the strong velocity gradients near the moving lid and the flow singularities at the corners. In particular, the smoothness and continuity of the streamlines in the near-wall regions demonstrate that the proposed semi-implicit discrete schemes introduce negligible numerical dispersion or artifacts, successfully capturing the physical flow features consistent with the integral form of the incompressible Navier-Stokes equations.

Fig. 8 provides a detailed quantitative assessment of the velocity field by comparing the centerline profiles against the high-fidelity benchmark data from Ghia et al. The left and right panels in each subfigure correspond to the u -velocity along the vertical mid-plane ($x = 0.5$) and the v -velocity along the horizontal mid-plane ($y = 0.5$), respectively. As shown in Fig. 8a, the baseline AD-PINN achieves excellent agreement with the reference data, validating the convergence of the neural network. Crucially, the proposed meshless frameworks, LSMPS-PINN (Fig. 8b) and LSFVP-PINN (Fig. 8c), exhibit nearly identical accuracy. Both methods successfully capture the sharp velocity gradients near the moving lid ($y \rightarrow 1$) and the stationary walls, demonstrating their capability to resolve thin boundary layers without grid refinement.

Furthermore, the peak velocities and the locations of the extrema in both u and v profiles are predicted with high precision by the proposed schemes. The coincidence of the predicted curves with the discrete benchmark points confirms that the semi-implicit Type-B operators do not introduce significant dissipation or dispersion errors, even in regions with high shear rates. This quantitative agreement reinforces the reliability of the proposed methods for simulating incompressible flows with complex boundary conditions.

The comparison of velocity magnitude and pressure fields shown in Fig. 9 reveals consistent methodological trends in resolving this benchmark flow. All approaches capture the primary vortex structure and basic velocity-pressure coupling, but significant differences emerge in field smoothness and physical consistency. AD-PINN exhibits noticeable numerical diffusion in velocity gradients and pressure oscillations, reflecting challenges in handling strong nonlinearities via automatic differentiation. LSMPS-PINN shows improved gradient

resolution with reduced artifacts, while LSFVP-PINN demonstrates enhanced physical fidelity with sharp boundary layer capture and physically coherent pressure distributions. The enhanced conservation properties of LSFVP-PINN validate its advantage in maintaining numerical stability for complex flow structures, whereas the efficiency of LSMPS-PINN renders it an attractive option for large-scale parametric studies.

4. Conclusions

This study has established a unified meshless operator learning framework that effectively bridges traditional numerical methods with physics-informed deep learning. Through the systematic development and validation of two complementary architectures—LSMPS-PINN and LSFVP-PINN—we have demonstrated that shifting the derivative computation from automatic differentiation to precomputed discrete operators fundamentally resolves the efficiency-accuracy bottleneck in training PINNs.

The comprehensive benchmarks across the Burgers equation, diffusion equation, and Lid-Driven cavity flow reveal that the two proposed methods establish two choices in the computational landscape:

- (1) LSMPS-PINN serves as the efficiency-oriented solution. By leveraging center-based strong-form operators with maximal sparsity, it achieves the fastest training throughput (reducing computational time by approximately 67 % compared to AD-PINN), making it ideal for rapid prototyping and large-scale parametric studies where speed is paramount.
- (2) LSFVP-PINN serves as the precision-oriented solution. By integrating Type-B LSMPS operators within an integral finite volume framework, it converts the governing PDEs into flux-based formulations. This dimensionality reduction in derivative order—from second-order Laplacian to first-order flux integration—delivers superior physical fidelity and numerical stability, achieving the lowest error rates at the cost of a moderate increase in operator complexity.

The core innovation lies in the flexible integration of least-squares approximations. While the center-based formulation (Type-A) prioritizes operator simplicity, the face-centered flux reconstruction (Type-B) in LSFVP enables simultaneous approximation of function values and derivatives while providing significantly improved local flux-balance properties compared to pure strong-form methods. This duality allows researchers to select the optimal solver architecture based on specific constraints: LSMPS-PINN for computational budget efficiency and LSFVP-PINN for rigorous error minimization via integral-based reduction.

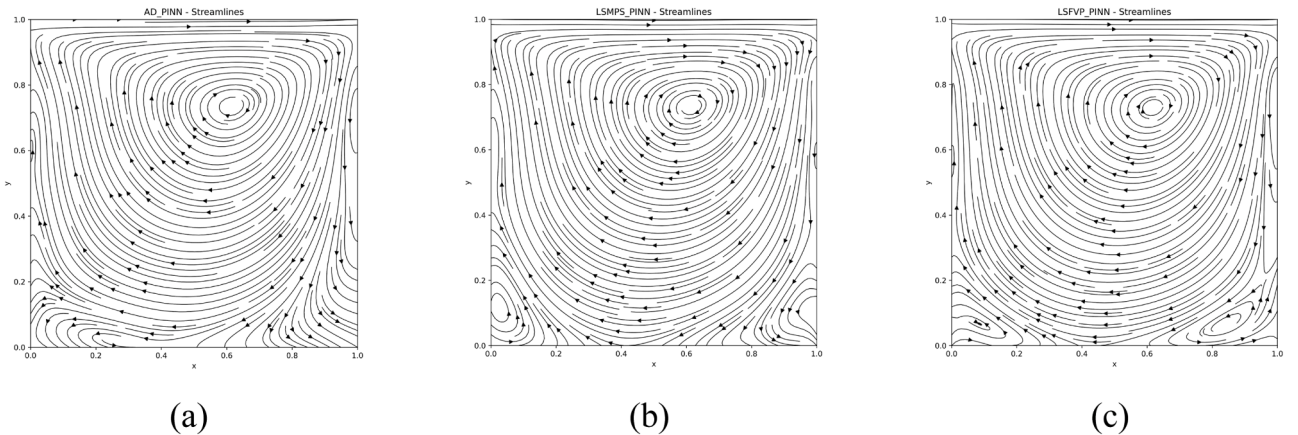


Fig. 7. Streamline patterns for (a) AD-PINN, (b) LSMPS-PINN, and (c) LSFVP-PINN methodologies.

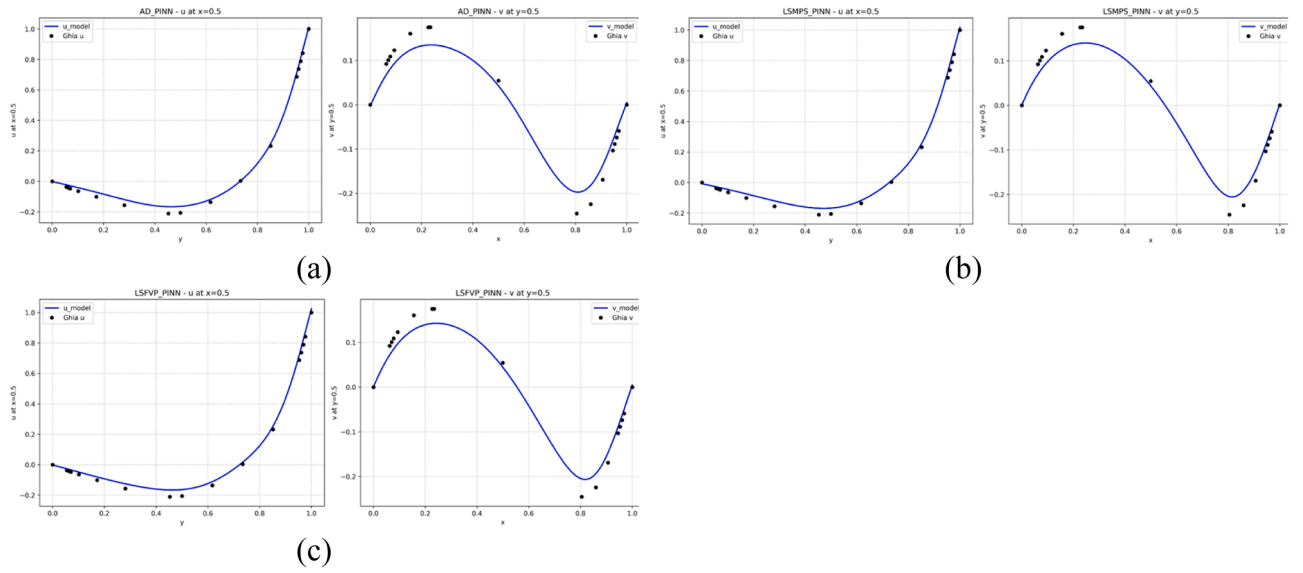


Fig. 8. Quantitative comparison of velocity profiles for the lid-driven cavity flow at $Re = 100$. The plots display the u -velocity along the vertical centerline ($x = 0.5$) and the v -velocity along the horizontal centerline ($y = 0.5$) for (a) the AD-PINN, (b) the LSMPS-PINN, and (c) the LSFVP-PINN frameworks.

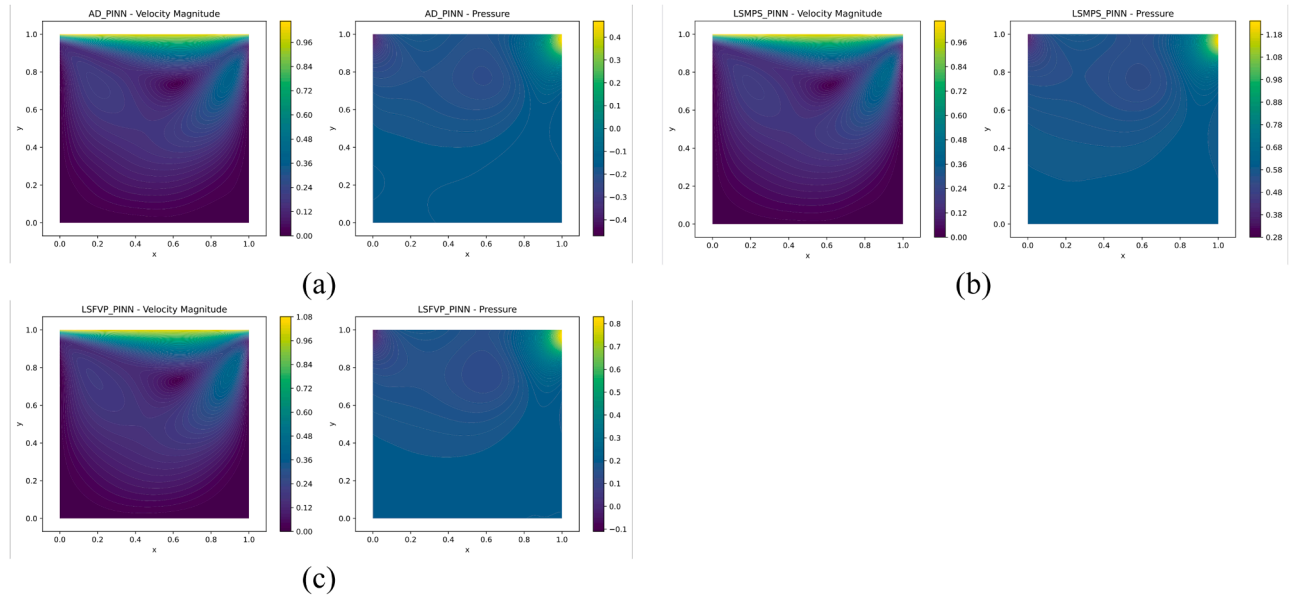


Fig. 9. Comparative velocity magnitude (left) and pressure (right) fields for lid-driven cavity flow at $Re=100$ for (a) the AD-PINN, (b) the LSMPS-PINN, and (c) the LSFVP-PINN frameworks.

Several practical considerations merit attention for implementation. Parameter selection, particularly regarding the influence radius and scaling factors, requires careful tuning to balance conditioning and locality. Furthermore, while LSFVP-PINN offers enhanced accuracy, its dense operator assembly for flux reconstruction incurs a higher memory footprint compared to the lightweight LSMPS-PINN, representing a necessary trade-off for conservation fidelity.

Future research directions should prioritize extending this unified framework to three-dimensional domains with adaptive particle refinement strategies. Additionally, exploring the application of the integral-based LSFVP formulation to shock-wave problems and coupled multi-physics systems, where conservation properties are critical, remains a promising pathway for advancing scientific machine learning.

CRediT authorship contribution statement

Zhen Zhang: Writing – original draft, Validation, Investigation. **Xiaoxing Liu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Shunhua Chen:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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